

Finding Inverses with Domain Restrictions

Finding the inverse and stating its domain, restricting a quadratic to one branch before inverting, and verifying an inverse pair by composition

Directions.

- A function has an inverse function only if it is **one-to-one**. If it is not, restrict the domain first and say which branch you kept.
- **An inverse is not finished until its domain is stated.** The domain of f^{-1} is the range of f , and the range of f^{-1} is the (restricted) domain of f .
- Method: write $y = f(x)$, swap x and y , solve for y , then choose the sign or branch that matches the restriction you kept.
- Check every inverse by composing: $f(f^{-1}(x))$ must simplify to x , or by testing one value both ways.

- 1.** Let $f(x) = 3x - 7$, with domain all real numbers.
(a) Find a formula for $f^{-1}(x)$. (b) Compute $f^{-1}(8)$.

Answer: _____

- 2.** Let $f(x) = \sqrt{x - 4}$, whose domain is $x \geq 4$.
(a) Find a formula for $f^{-1}(x)$ and state its domain.
(b) Compute $f^{-1}(3)$, then check your formula by computing $f(13)$.

Answer: _____

3. Let $f(x) = x^2 - 6x + 5$, restricted to $x \geq 3$ so that it is one-to-one.

- (a) Find a formula for $f^{-1}(x)$ and state its domain.
- (b) Compute $f^{-1}(12)$, then check by computing $f(7)$.

Answer: _____

4. Let $g(x) = \frac{2x+1}{x-3}$, with $x \neq 3$.

- (a) Verify by composition that $g^{-1}(x) = \frac{3x+1}{x-2}$.
- (b) State the value that must be excluded from the domain of g^{-1} , and say which feature of g it comes from.
- (c) Compute $g^{-1}(3)$.

Answer: _____

5. Let $f(x) = (x + 2)^2 - 5$, restricted to $x \leq -2$.

(a) Find a formula for $f^{-1}(x)$ and state its domain. Say clearly which branch of the parabola you kept and how that fixed the sign.

(b) Compute $f^{-1}(11)$, then check by computing $f(-6)$.

Answer: _____

6. Let $f(x) = 9 - x^2$, restricted to $x \leq 0$.

(a) Find a formula for $f^{-1}(x)$ and state both its domain and its range.

(b) Compute $f^{-1}(5)$, then check by computing $f(-2)$.

Answer: _____

- 7.** Let $f(x) = \sqrt{2x - 6} + 1$, whose domain is $x \geq 3$.
(a) Find a formula for $f^{-1}(x)$ and state its domain.
(b) Compute $f^{-1}(5)$, then check by computing $f(11)$.

Answer: _____

- 8.** Priya claims that the inverse of $f(x) = (x + 4)^2$, restricted to $x \geq -4$, is given by the rule $x^2 - 4$. Testing her rule at $x = 9$ gives 77.
(a) Compute the true value of $f^{-1}(9)$, and check it by computing $f(-1)$.
(b) Give the correct formula for $f^{-1}(x)$, state its domain, and say in one sentence what Priya reversed.

Answer: _____

9. Let $f(x) = x^2 + 8x + 12$.

- (a) Find the largest interval of the form $x \geq c$ on which f is one-to-one, and give the value of c .
- (b) On that restricted domain, find a formula for $f^{-1}(x)$ and state its domain.
- (c) Compute $f^{-1}(5)$, then check by computing $f(-1)$.

Answer: _____

10. Explain, in complete sentences, why a function must be one-to-one before it can have an inverse function. Use $f(x) = x^2 - 4$ on all real numbers as your example: compute $f(3)$ and $f(-3)$, explain exactly what goes wrong when you try to define $f^{-1}(5)$, and describe one domain restriction that repairs the problem — including what the inverse's domain and range become once you have made it.

Answer: _____